

PHYSICS • BIOLOGY • THERMODYNAMICS

From Schrödinger to Press to Kleiber

The characteristic scale of living matter

By Lee McCulloch-James

Press's calculation³ can be read as part of a larger question. Once matter is capable of the weak, reversible chemistry required by life, what sets the characteristic size and energetic rate of an organism?

The chemical scale

The natural atomic energy scale is the Rydberg, built from the Bohr radius a_0 , the characteristic size of a hydrogen atom's ground state,

$$a_0 = \frac{\hbar^2}{m_e e^2} \simeq 0.53 \text{ Å}, \quad E_{\text{Ry}} = \frac{e^2}{2a_0} = 1 \text{ Ry} \simeq 13.6 \text{ eV}.$$

PHYSICAL FOOTHOLD

Why is this “natural”? Nuclear physics has a familiar version of this move: the liquid-drop model sets a characteristic nuclear energy scale ($\sim 8 \text{ MeV}$ per nucleon) by balancing a volume term, a surface term ($\propto A^{2/3}$) and a Coulomb term against each other. The atom has its own balancing act, between quantum kinetic energy and electrostatic attraction.

An electron confined to a radius r carries kinetic energy $\sim \hbar^2/2m_e r^2$ (the uncertainty principle: squeeze it in space and its momentum, hence kinetic energy, must rise) and Coulomb energy $-e^2/r$. Minimising

$$E(r) \sim \frac{\hbar^2}{2m_e r^2} - \frac{e^2}{r}$$

over r gives back exactly the Bohr radius a_0 stated above, and substituting back gives $E_{\text{Ry}} = e^2/2a_0 = m_e e^4/2\hbar^2 \simeq 13.6 \text{ eV}$. That number is not an abstract textbook constant, it is exactly the ionization energy of hydrogen's ground state: the energy to strip the one electron off the one proton. Every other energy in this argument is measured against that benchmark. Here is roughly where a few familiar bond strengths sit on the same ladder⁵:

Interaction	Energy	in Ry
Hydrogen ionization energy	13.6 eV	1
Covalent bond (C–C, C–H)	$\sim 4 \text{ eV}$	~ 0.3
Hydrogen bond (DNA base pair)	0.1–0.3 eV	$\sim 0.01\text{--}0.02$
Van der Waals contact	0.01–0.06 eV	$\sim 10^{-3}$
Thermal energy $k_B T$ (300 K)	0.026 eV	~ 0.002
Press's $\epsilon \text{ Ry}$	0.041 eV	0.003

The double helix's hydrogen bonds, and the van der Waals contacts packed around it, sit only a few $k_B T$ above the thermal noise floor. That is exactly why they can be broken and reformed by ordinary thermal jostling, unzipped for replication, refolded into a new shape, without the large activation energy a covalent bond would demand. Press's $\epsilon \text{ Ry}$ lands in that same narrow band.

Press introduces a dimensionless parameter ϵ to represent the much smaller energy scale relevant to complex molecular chemistry,

$$E_{\text{chem}} \sim \epsilon \text{ Ry}, \quad \epsilon \simeq 0.003.$$

Thus

$$\epsilon \text{ Ry} \simeq 0.041 \text{ eV},$$

which, as the table above shows, is squarely in the range of a weak hydrogen bond. This is already suggestive in that at ordinary biological temperatures,

$$k_B T \simeq 0.026 \text{ eV} \quad (T \simeq 300 \text{ K}),$$

so that

$$\frac{\epsilon \text{ Ry}}{k_B T} \sim 1.$$

The operative scale of weak molecular chemistry therefore lies close to the scale of thermal agitation.

This gives a natural continuation of the problem posed by Schrödinger in *What Is Life?* (1944)¹. Molecular structures must be sufficiently stable to preserve information, while many of the interactions responsible for recognition, folding, binding and release must remain thermally accessible. Schematically,

$$E_{\text{weak}} \sim \epsilon \text{ Ry} \sim \mathcal{O}(k_B T).$$

The small parameter ϵ may therefore be understood as measuring the separation between the atomic electronic scale and the softer molecular scale on which living machinery operates.

Press: how large can chemically bound matter become?

Press then combines this chemical scale with gravity, through three physical requirements he places on a body built like ours: it is made of ordinary molecules, bound at $\sim \epsilon \text{ Ry}$; it lives on a planet whose atmosphere neither boils away into space nor liquefies underfoot; and it is as large as it can be while still surviving an ordinary fall.

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Where the formula actually comes from. The first ingredient is density. Atoms pack shoulder to shoulder at a spacing of about one atomic *diameter*, not one radius, so ordinary condensed matter has a characteristic density

$$\rho_0 \sim \frac{m_p}{(2a_0)^3} \simeq 1.4 \text{ g/cm}^3,$$

already close to water and rock. This is why $2a_0$, the Bohr *diameter*, rather than a_0 itself, is the length that ends up in the final formula: it is the packing scale, not the orbital radius.

The second ingredient is the “don’t break” condition. Falling from a height L_H delivers kinetic energy $\sim M_H L_H g$, where $g = GM_\oplus/R_\oplus^2$ is the local surface gravity. What resists that fall is the number of atomic bonds spanning the body’s narrowest cross-section, roughly $(M_H/m_p)^{2/3}$ of them, each worth $\epsilon \text{ Ry}$. Setting the fall energy equal to the total bond energy,

$$M_H L_H g \sim \epsilon \text{ Ry} (M_H/m_p)^{2/3},$$

and then eliminating g , M_H , $M_\oplus$ and $R_\oplus$ using ρ_0 and the atmosphere condition (which, as a byproduct, also predicts the Earth’s own mass and radius purely from ϵ and fundamental constants³) leaves L_H as a function of ϵ and the fundamental constants alone. Carrying that algebra through is what produces the specific combination $\epsilon^{1/4}(2a_0)(e^2/Gm_p^2)^{1/4}$: the $2a_0$ supplies the atomic length scale, and the dimensionless ratio e^2/Gm_p^2 supplies the leverage that gravity and electromagnetism exert against one another.

His estimate gives a characteristic organism size

$$L_H \sim \epsilon^{1/4}(2a_0) \left(\frac{e^2}{Gm_p^2} \right)^{1/4},$$

or numerically,

$$L_H \simeq 2.6 \left(\frac{\epsilon}{0.003} \right)^{1/4} \text{ cm.}$$

The weak dependence

$$L_H \propto \epsilon^{1/4}$$

is important as even considerable uncertainty in the detailed chemistry produces only a modest change in the characteristic length.

PHYSICAL FOOTHOLD

What is e^2/Gm_p^2 ? It is a pure number: the ratio of the electrostatic repulsion between two protons to their mutual gravitational attraction,

$$\frac{e^2/r^2}{Gm_p^2/r^2} = \frac{e^2}{Gm_p^2} \simeq 2.3 \times 10^{36}.$$

This is the same enormous ratio behind Dirac's large-number hypothesis⁶ and the older puzzle, going back to Weyl and Eddington, of why gravity is so much feebler than electromagnetism. Press's L_H takes the *fourth root* of this number, and a fourth root is what tames a colossal 10^{36} down to something of order unity: $(10^{36})^{1/4} \sim 10^9$, multiplied onto a Bohr radius of $\sim 10^{-8}$ cm, lands at a few cm. The same large number that separates gravity from electromagnetism throughout physics is, after that fourth root, a large part of why organisms come out at the scale of centimetres rather than microns or kilometres.

That same imbalance is also, in a looser sense, part of why stars shine steadily for billions of years instead of collapsing on the spot: gravitational binding energy per particle is minuscule next to nuclear or chemical binding energy per particle for any object up to stellar mass, precisely because gravity is $\sim 10^{36}$ times feebler than electromagnetism at the particle level. It is worth keeping that separate, though, from a different "why so slow" number that also governs the Sun: the proton-proton chain's first step, $p + p \rightarrow d + e^+ + \nu$, proceeds through the weak interaction, and it is the weak force's own tiny coupling, not the gravity-versus-electromagnetism ratio, that makes a given proton wait, on average, on the order of billions of years before it fuses. Gravity's weakness is why the Sun doesn't implode; the weak interaction's is a separate reason it doesn't detonate. Both are enormous numbers, both make stars behave gently, but they are not the same number, and it is easy to reach for one when the other is meant.

Press is therefore asking, in effect: given chemically bound matter operating in the thermal window required for complex molecular processes, how large can an object made from that matter become before gravity becomes mechanically important?

Kleiber: living matter is not homogeneously distributed

There is another scaling problem once the object is alive. A geometrically homogeneous body has

$$M \propto L^3, \quad A \propto L^2.$$

If its energetic exchange with the environment were controlled simply by its external surface, one would expect

$$P \propto A \propto L^2$$

and hence

$$P \propto M^{2/3}.$$

This is the natural scaling for an approximately homogeneous radiating body: a three-dimensional bulk exchanging energy through a two-dimensional boundary.

Empirically though, whole-organism metabolic rate is described instead by Kleiber's law (1932)²,

$$P_{\text{metabolic}} \propto M^{3/4}.$$

The exponent of 3/4 correction to the expected Stefan–Boltzmann surface-area law has a precise origin that remains a matter of modelling and empirical discussion. One influential explanation, associated with hierarchical resource-distribution models such as West, Brown and Enquist's fractal network theory (1997)⁴, is that an organism is not a homogeneous lump whose relevant interface is merely its external surface.

Living matter is threaded by internal transport and exchange structures: vascular networks, capillaries, respiratory surfaces, cell membranes and mitochondrial machinery. Metabolism occurs throughout the volume, while matter and free energy must continually be transported through this distributed, approximately fractal architecture.

PHYSICAL FOOTHOLD

Where does the 3/4 actually come from? Kleiber's law is tested over an enormous range, from a mouse (~ 20 g) to a blue whale ($\sim 10^5$ kg), about seven orders of magnitude in mass. Over that range $M^{2/3}$ and $M^{3/4}$ predictions pull apart enough to be told apart on a log-log plot, and the data track the 3/4 line. West, Brown and Enquist's own way of putting it is that a fractal, space-filling transport network makes an organism behave, for the purposes of resource delivery, as though it occupies one extra effective dimension: not the 3 of ordinary bulk matter, but not a full 4 either.

The geometrical picture therefore changes from: three-dimensional bulk $\rightarrow$ two-dimensional exterior to something closer to:

three-dimensional living matter $\rightarrow$ hierarchical internal transport network $\rightarrow$ distributed molecular interfaces.

This is the point at which the physics of a passive body gives way to the physics of organised living matter.

The metabolic quarter power

Kleiber scaling immediately gives

$$\frac{P}{M} \propto M^{-1/4}.$$

The metabolic power available per unit mass therefore decreases with organism size. A gram of a small animal processes energy more rapidly than a gram of a large animal. This can be flipped suggestively:

$$\frac{M}{P} \propto M^{1/4}.$$

The appearance of a quarter power is striking beside Press's result $L_H \propto \epsilon^{1/4}$. Their coexistence is an invitation to ask whether Press's mechanical estimate and Kleiber's metabolic scaling can be placed within a common hierarchy.

Two chemical energy scales

There is also an important distinction between the energy that *drives* metabolism and the weak interactions through which much of its molecular machinery operates.

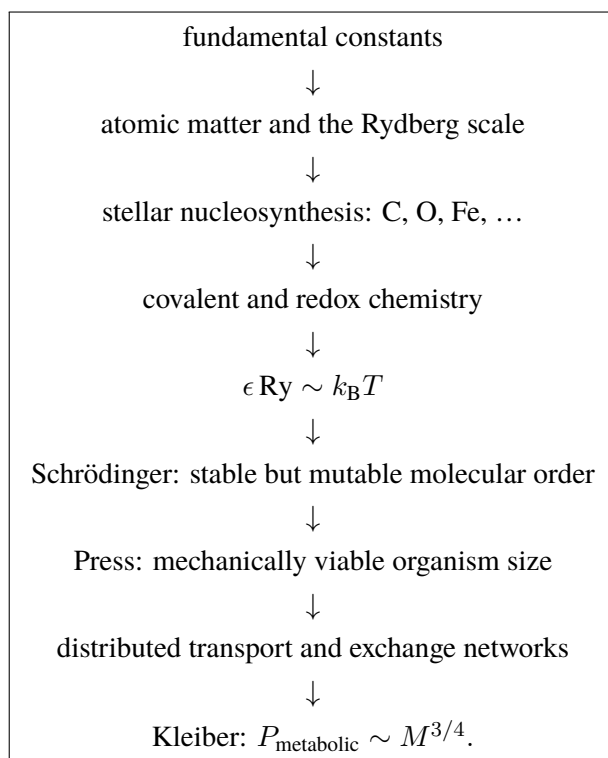
$$\begin{array}{ll} \text{redox and covalent chemistry} & \Delta G_{\text{chem}} \sim \text{eV} \\ \downarrow & \\ \text{weak molecular machinery} & E_{\text{weak}} \sim \epsilon \text{ Ry} \sim k_B T. \end{array}$$

Elements such as carbon, oxygen and iron make possible a rich hierarchy of covalent, redox and catalytic chemistry. In particular, oxygen chemistry and transition-metal centres provide routes by which relatively large chemical free-energy differences can be coupled to molecular processes operating close to the thermal scale.

Thus a living organism contains a hierarchy of energies: larger chemical free-energy changes continually maintain a vast network of weaker, thermally accessible molecular interactions away from equilibrium.

A possible hierarchy of living scale

The argument can therefore be organised as a progression:



The progression poses three increasingly macroscopic questions.

Schrödinger asks how molecular matter can maintain organised states in a thermal environment.

Press asks what characteristic size an organism made from such matter can attain under gravity.

Kleiber scaling asks how the resulting extended organism supplies and dissipates the free energy required to keep its molecular machinery operating throughout its volume.

WHAT'S STILL OPEN

Press's original Stefan–Boltzmann argument treats the organism through its external radiating area. A living organism, however, processes energy through a hierarchical internal architecture. The open mathematical question is whether Press's surface-flux step, $P \propto M^{2/3}$, can be replaced by a metabolic scaling law, $P \sim M^{3/4}$, and whether doing so produces a characteristic timescale that is less sensitive to the phenomenological parameter ϵ than Press's original result, $t_H \propto \epsilon^{-2.75}$. That calculation would provide a direct mathematical splice between Schrödinger's thermodynamic window, Press's characteristic size, and Kleiber's metabolic scaling, rather than the three simply being placed side by side.

KEY TERMS

Rydberg (Ry). The natural atomic unit of binding energy, $e^2/2a_0 \simeq 13.6$ eV, set by the Bohr radius a_0 ; equal to the ionization energy of ground-state hydrogen.

Bohr radius (a_0). The characteristic size of a hydrogen atom's ground state, $a_0 = \hbar^2/m_e e^2 \simeq 0.53$ Å, set by balancing an electron's quantum kinetic energy against its Coulomb attraction to the proton.

Ionization energy. The energy needed to remove an electron entirely from an atom or molecule; for ground-state hydrogen this equals exactly 1 Ry.

ϵ (**Press's chemical parameter**). A dimensionless factor, $\epsilon \simeq 0.003$, representing the fraction of a Rydberg characteristic of weak molecular (as opposed to atomic) binding energies.

Hydrogen bond. A relatively weak (~ 0.1 – 0.3 eV) electrostatic attraction, for example between base pairs in DNA, strong enough to hold structure but weak enough to be broken by ordinary thermal motion.

Van der Waals interaction. A very weak (~ 0.01 – 0.06 eV), short-range attraction between neutral atoms or molecules, arising from fluctuating electric dipoles.

Activation energy. The energy barrier that must be supplied, often by a thermal fluctuation, before a bond can break or a reaction can proceed.

Kleiber's law. The empirical observation that resting metabolic rate scales as roughly the $3/4$ power of body mass across a wide range of organisms (Kleiber, 1932).

Fractal network model (WBE). The West–Brown–Enquist proposal that Kleiber's $3/4$ exponent arises from space-filling, hierarchically branching internal transport networks rather than from external surface area alone (West, Brown, and Enquist, 1997).

Homogeneous-body scaling. The naive expectation, $P \propto M^{2/3}$, for a body that exchanges energy with its environment only through its outer surface.

FURTHER READING

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- [4] West, G. B., Brown, J. H., and Enquist, B. J. (1997). A General Model for the Origin of Allometric Scaling Laws in Biology. *Science* 276(5309), 122–126.
- [5] Phillips, R., Kondev, J., Theriot, J., and Garcia, H. G. (2012). *Physical Biology of the Cell*, 2nd ed. Garland Science. (Source for the energy-scale comparison table: hydrogen bonds, van der Waals contacts, and covalent bonds measured against $k_B T$.)
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